HoloAssist – “Futurino UR3eno”

Client: Augustine Le

Coach: Tan Huynh

Group Name: Thu_9_12_00

Team Members:

Full Name	Student ID	Expertise
John Chen	24837332	ROS2 system architecture, robot autonomy tooling, motion planning and control integration, reactive Cartesian velocity control, and perception–control interface design.
Sebastian Baudille	24804940	Autonomous systems integration, embedded systems development.
Oliver Lau	24770051	Embedded systems and electronics, PCB design (Altium), STM32 microcontrollers, hardware interfacing, sensor integration, and system-level electrical implementation.
Nicholas Sabatta	24796955	Robotics system development, robot hardware setup, mechanical integration, experimental validation, and collaborative task testing.

Overview

1.1 Project Description

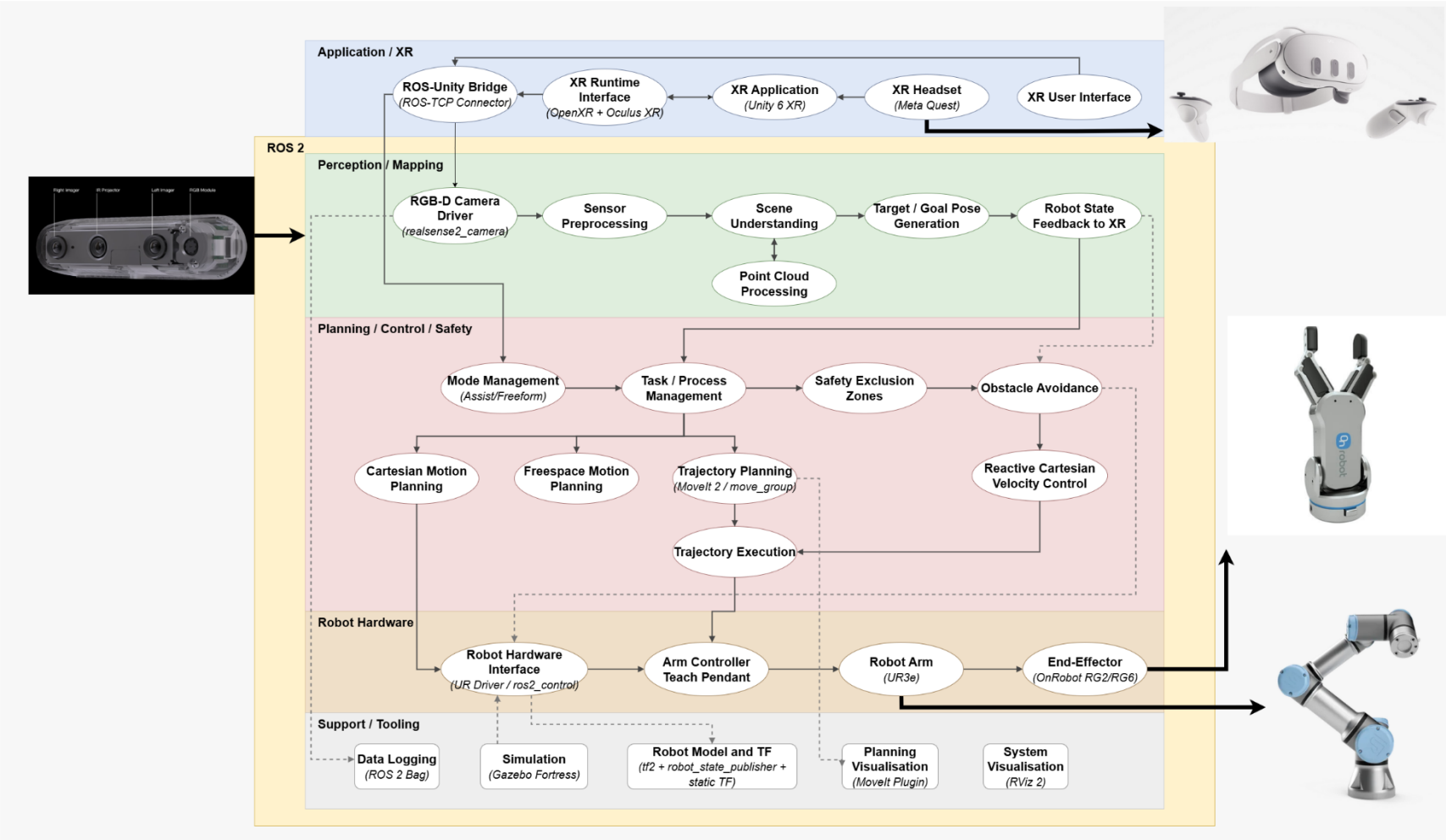
HoloAssist is an XR-based teleoperation system that enables a user to remotely control a UR3e collaborative robot while viewing the robot and its surrounding workspace through an augmented reality headset.

The system provides:

- Live visualisation of the robot model and end-effector pose
- Real-time depth-based spatial awareness of the workspace
- Basic dynamic safety boundaries derived from perception
- Assisted and free-form teleoperation modes

The system is designed for a hypothetical industry scenario in which a user must operate a robot from outside a hazardous workspace. The focus of this project is to improve operator spatial awareness and safety through XR visualisation and constrained control within a hybrid remote-control and autonomous robotic system.

1.2 High-level System Diagram



Aims

1.3 Project Deliverables

The primary aim of this project is to design, implement, and evaluate a constraint-aware XR teleoperation framework for collaborative robotics. The following deliverables will be produced:

ID	Deliverables	Description
D1	End-effector Control	A reactive Cartesian velocity control module enabling safe teleoperation of a collaborative robot.
D2	Perception	A perception pipeline integrating an RGB-D camera to generate and process real-time point cloud data.
D3	Obstacle Avoidance	A dynamic exclusion zone generation system derived from live perception data to enforce workspace safety constraints.
D4	XR/AR Human-Robot Interface (HRI)	An XR interface (Meta Quest 2/3) capable of: – Visualising the robot model and end-effector pose – Rendering live point clouds of the workspace – Displaying predicted and executed trajectories – Displaying dynamic exclusion zones and safety states
D5	High-level application control	A mode management framework supporting: – Freeform teleoperation – Assisted constraint-aware teleoperation – Teleoperation recording and replay (training mode)
D6	Demonstration	An integrated system demonstration performing a validated collaborative task (e.g. tool handover or alignment assistance).
D7	Testing	A quantitative evaluation framework measuring task performance, safety interventions, and system responsiveness.
D8	Documentation	A complete system architecture document and risk assessment aligned with studio requirements.

1.4 Key Milestones

The project will be structured across sprint-based milestones. Each milestone has a set of project tasks.

1.4.1 Sprint 1 (Weeks 0–3): Scoping and Architecture

Project Task	Description
Project contract	Finalised project contract and subsystem definitions.
Risk assessment	Risk assessment completed.
System architecture	System architecture diagram defining subsystem interfaces.
Test stack + template	Initial robot bring-up and verification of the communication pipeline.

1.4.2 Sprint 2 (Weeks 4–7): Core System Implementation

Project Task	Description
End-effector control	Reactive velocity control implemented and validated.
Perception stack	RGB-D perception pipeline publishing usable point cloud data.
XR/AR Interface	XR interface displaying robot state and controller input.
Simulation environment	Basic workspace with boundaries and interactable items

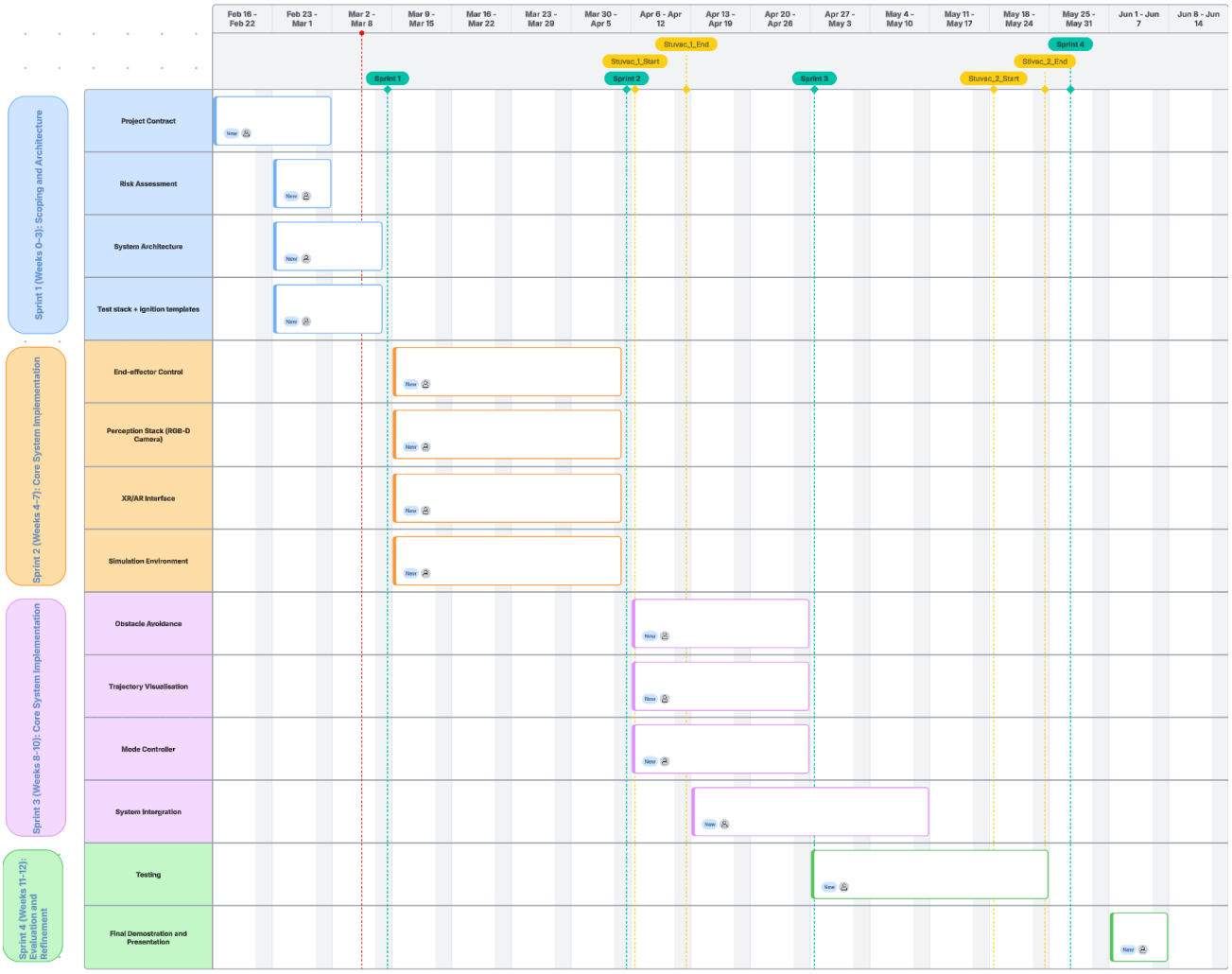
1.4.3 Sprint 3 (Weeks 8–10): Integration and Constraint Enforcement

Project Task	Description
Obstacle avoidance	Dynamic no-go zones are implemented and integrated with the control layer.
Trajectory visualisation	Trajectory preview visualisation is operational in XR.
Mode controller	Mode switching between freeform and assisted control was validated.
System integration	End-to-end system integration achieved.

1.4.4 Sprint 4 (Weeks 11–12): Evaluation and Refinement

Project Task	Description
Testing	System robustness and safety were validated through repeated trials.
Final demonstration and presentation	Final demonstration and project handover preparation.

Time-Action Plan (Gantt Chart)



1.5 Learning Goals

1.5.1 Team Learning Goals

As a team, we aim to:

- Apply systems engineering principles to design, integrate, and validate a multi-subsystem robotics solution.
- Develop and evaluate a robotics path planning and control framework within a real-world collaborative context.
- Strengthen interdisciplinary collaboration through clearly defined subsystem ownership and interface agreements.
- Improve technical communication through sprint reviews, documentation, and reflective critique.
- Develop resilience in debugging and integration of complex robotic systems.

1.5.2 Individual Learning Goals

Each team member aims to deepen expertise within their subsystem while strengthening integration awareness:

- Develop advanced understanding of reactive motion control and constraint-based robotics.
- Gain practical experience integrating perception data into real-time control frameworks.
- Improve XR-based interface design for robotics applications.
- Strengthen hardware–software integration and safety-critical system thinking.
- Enhance professional capabilities in teamwork, reflection, and iterative design improvement.

Resources

The successful completion of this project is incumbent on access to robotics hardware, perception systems, XR equipment, fabrication facilities, and supporting software frameworks. The required resources are outlined below.

1.6 Hardware and Software (Laboratory Provided)

ID	Hardware	Quantity	Description
H1	UR3e Collaborative Robot	1	6-DOF Universal Robots UR3e collaborative manipulator used as the primary robotic platform for teleoperation, velocity control implementation, and collaborative task execution.
H2	Mobile Robot Workstation	1	Trolley-mounted UR3e setup, including base frame and workspace surface, enabling flexible repositioning within the lab environment while maintaining safety compliance.
H3	UR3e Teach Pendant	1	Official Universal Robots teach pendant used for robot configuration, safety setup, emergency stops, manual jogging, and recovery operations during development and testing.
H4	Intel RealSense RGB-D Camera	1	RGB-D depth camera used for real-time point cloud generation, workspace sensing, and dynamic safety boundary estimation.
H5	Camera Mount / Tripod	1	Adjustable tripod or mounting bracket for stable positioning of the RGB-D camera in either eye-in-hand or eye-to-hand configuration.
H6	Parallel Gripper	1	Two-finger end-effector for executing simple collaborative manipulation tasks (e.g., object alignment, placement, or handover demonstrations).
H7	Workstation PC	1	High-performance desktop workstation running ROS 2, perception processing, control stack, and XR communication bridge.
H8	Meta Quest 2 Headset	1	XR headset used for augmented reality visualisation of robot pose, point clouds, trajectories, and safety boundaries.
H9	USB-C to Ethernet Adapter	1	An adapter that enables stable wired network connectivity for the Meta Quest headset, reducing wireless latency and improving communication reliability.
H10	Ethernet Cables	2–3	CAT6 Ethernet cables for direct communication between workstation, UR3e controller, networking hardware, and XR device.

ID	Software	Description
S1	Ubuntu 22.04 LTS & ROS 2 Humble	Core operating system and robotics middleware used for communication between perception, control, and XR subsystems.
S2	MoveIt 2	Motion planning framework used for robot modelling, kinematics, and optional trajectory validation.
S3	Universal Robots ROS2 Driver	Interface enabling communication between the UR3e robot controller and the ROS 2 environment.
S4	RViz2	3D visualisation tool used for debugging robot states, transforms, trajectories, and point cloud data.
S5	RealSense ROS2 Packages	Drivers for acquiring RGB-D data and generating point clouds from the Intel RealSense camera.
S6	Gazebo	A simulation tool to support the visualisation and testing of the system virtually without hardware.
S7	Unity 6	A game engine is used to develop the XR interface and visualise robot state and workspace information.
S8	Unity XR Interaction Toolkit	Toolkit used within Unity to handle headset tracking and controller input for XR interaction.
S9	ROS–Unity Communication Bridge	Middleware enabling communication between ROS 2 nodes and the Unity XR environment.
S10	tf2	ROS transform library used to manage coordinate frames between the robot, camera, and XR reference systems.
S11	PCL (Point Cloud Library)	Used for filtering and processing real-time point cloud data.
S12	OpenCV	Image processing library for filtering and validating RGB-D image data.
S13	rosbag2	A logging tool that supports the record of system data for testing, debugging, and evaluation.
S14	Git and GitHub	Version control platform used for collaborative development, code management, and subsystem integration.

1.7 Modifications to Laboratory Hardware

ID	Modification	Description	Purpose
M1	Custom Tool Holder / Gripper Adapter	Design and fabrication of a task-specific attachment compatible with the UR3e parallel gripper.	Enables execution of the selected demonstration task and mimics a simplified industry-relevant application.
M2	RGB-D Camera Mounting Bracket	Design and fabrication of a secure mounting solution for the Intel RealSense camera. Mount location may be on the robot base, workspace frame, or end-effector, depending on chosen configuration (eye-in-hand or eye-to-hand).	Ensures stable depth sensing and accurate spatial alignment for perception and safety boundary generation.

All modifications will comply with laboratory safety procedures and be reviewed with the lab manager before being added to the existing lab hardware.

1.8 Hardware and Software (Team-Sourced)

ID	Item	Type	Description
TS1	Meta Quest 2	Hardware	XR headset used for augmented reality visualisation and teleoperation interface testing.
TS2	Meta Quest 3	Hardware	An optional upgraded XR headset is used to improve tracking stability and display performance during testing and demonstrations.
TS3	Unity Hub & Unity 6	Software	Development environment for building and managing the XR interface and spatial rendering pipeline.
TS4	ROS–Unity Communication Bridge	Software	Middleware enabling bidirectional communication between ROS 2 nodes and the Unity XR environment (e.g. ROS-TCP-Connector or equivalent).
TS5	Dedicated Networking Hardware (Optional)	Hardware	An optional router or wired networking setup to improve communication stability and reduce latency between the XR device and the control workstation.

System Evaluation

The fully integrated HoloAssist system enables an operator wearing an XR headset to teleoperate a UR3/UR3e collaborative robot using reactive Cartesian end-effector velocity control. The operator views a live spatial overlay of robot state, trajectories, point clouds, and dynamic exclusion zones derived from an RGB-D perception stack. The system supports both freeform and assisted teleoperation modes, together with telemetry and visual feedback.

Evaluation Level	Description/Tasks
Pass (MVP)	• XR headset displays a stable robot model and end-effector pose, and robot state updates reliably in XR during motion. • The operator can teleoperate the robot using reactive velocity control, and the system can stop safely if communication is lost or teleoperation is disabled. • Depth camera monitors workspace independent of robot and publishes to Desktop UI on a separate device. • Operator can observe pose markers/robot transforms in XR or desktop visualisation.
Credit	• With an assisted mode enabled, the robot will autonomously maintain a safe working distance from the human via a safety boundary by planning. subsequent trajectories to be replanned using a collision box in MoveIt. • A basic teleoperation mode is operational through XR input, with task completion achievable in repeated trials.
Distinction	• Safety boundary dynamically updates from live depth data at 5hz with updates gated to account for object settling. • Multiple teleoperation modes are supported, such as end-effector control and manual joint manipulation. • Demonstration task completed consistently under a defined time threshold. • Passthrough or spatial marker visualisation is working.
High-Distinction	• Constraint-aware velocity modulation based on proximity distance. • Assisted behaviour adapts based on proximity bands, applying progressively more conservative MoveIt safety regions as the operator gets closer to obstacles in the robot's environment (e.g. table, person). • Quantitative comparison between freeform and assisted modes. • UI in mixed reality can visualise and debug the depth camera feed (maybe by painting areas of importance)

Subsystem Evaluation

Subsystem 1 - **Perception and Mapping**	**Lead by: John**
P	Depth camera monitors the workspace independently of the robot and publishes to a desktop UI on a separate device.
C	Perception output supports assisted mode by providing workspace information used to define a basic safety boundary for MoveIt planning.
D	Safety boundary updates dynamically from live depth data at approximately 5 Hz, with updates gated to account for object settling and reduce noisy changes.
HD	Assisted behaviour adapts based on proximity bands derived from depth data, applying progressively more conservative safety regions as obstacles or the operator move closer to the robot workspace.
Perfect	Depth-based workspace monitoring remains stable and reliable during full-system demonstration, including under simulated disturbance or transient sensing noise.
Subsystem 2 - **Motion Planning and Control**	**Lead by: Oliver**
P	Reactive Cartesian velocity control is operational, and the system can stop safely if communication is lost or teleoperation is disabled.
C	With assisted mode enabled, subsequent trajectories are replanned in MoveIt using a predefined collision boundary to maintain a safe working distance from the human.
D	Multiple teleoperation modes are supported, including end-effector control and manual joint manipulation.
HD	Constraint-aware velocity modulation is applied based on proximity distance, with motion becoming more conservative as the robot approaches obstacles or human-adjacent regions.
Perfect	Motion planning and execution remain smooth, responsive, and robust across teleoperation, assisted motion, and trajectory preview during the final integrated demonstration.

Subsystem 3 - **Interaction and Execution**	**Lead by: Nicholas**
P	XR headset displays a stable robot model and end-effector pose, and robot state updates reliably in XR during motion.
C	A basic teleoperation mode is operational through XR input, with task completion achievable in repeated trials.
D	The operator can switch between supported teleoperation modes, and passthrough or spatial marker visualisation is operational.
HD	Mixed reality UI supports visualisation and debugging of the depth camera feed, such as highlighting or painting regions of interest in the workspace.
Perfect	XR interaction is stable and low jitter, with smooth trajectory preview visualisation and a polished user experience suitable for public demonstration.
Subsystem 4 – **XR and Scene Visualisation**	**Lead by: Sebastian**
P	Operator can observe pose markers and robot transforms in XR or desktop visualisation.
C	System visualisation supports repeated demonstration trials by clearly presenting robot state, task progress, and safety context to the operator.
D	Demonstration task is completed consistently within a defined time threshold, supported by integrated XR scene feedback and visual cues.
HD	Visual outputs support quantitative comparison between freeform and assisted modes, including clear representation of active safety regions, control state, or performance overlays.
Perfect	Final XR scene integration is polished and showcase-ready, including stable trajectory preview visualisation in mixed reality and robust behaviour under simulated failure conditions.

Appendix

Appendix A

Other potential names: HoloMotion, VisionLayer, Spatial Robotics, OmniSpace, Telessist, TeleMotion, HoloLens